

S. No.	Key Skills or Technologies	Proposed Training Methods	Tentative Timelines	Expected Outcomes or Goals	Suitable For
1	Advanced Python Concepts (List, dict, Decorators, Generators, Asyncio, Type Hints)	Udemy / Educative / Real Python + internal code reviews	3 weeks	Solid understanding of advanced Python coding practices	Fresher
2	Django Advanced (MVT, ORM, Signals, Middleware, Caching, REST APIs with DRF)	Django Official Docs + Udemy (Django Mastery) + Mini Projects	4 weeks	Confident in building scalable, secure Django apps	All
3	FastAPI with Async & Pydantic / Flask	FastAPI Docs + Full Course (e.g., "FastAPI Complete Guide" on Udemy) / Flask Full Course	3 weeks	Able to develop async microservices and REST APIs	All
4	LangChain Essentials (Chains, Agents, Tools)	LangChain Official Docs + YouTube Tutorials + Example Apps	3 weeks	Build LLM-powered pipelines with LangChain	All
5	LangGraph for LLM Workflows	LangGraph Docs + Community Projects + Open Source Examples	2 weeks	Implement agent-based, stateful LLM workflows	All
6	Retrieval Augmented Generation (RAG) - Simple to Advanced	RAG Paper + YouTube (e.g., Pinecone, Cohere) + Custom App	4 weeks	Build robust RAG pipelines using vector DBs	All
7	Vector Databases (FAISS, Chroma, Weaviate)	Tutorials + LangChain Integration	2 weeks	Efficient chunking, embedding & retrieval for RAG	All

8	OpenAI, Anthropic, and HuggingFace APIs	Official Docs + Sample Projects	2 weeks	Use different LLMs for different use cases	All
9	Prompt Engineering	DeepLearning.AI Prompt Engg Course + LangChain usage	2 weeks	Write effective prompts for complex tasks	All
10	Dockerizing AI Apps (Docker + Docker Compose)	YouTube / Udemy + Hands-on deployment	2 weeks	Containerize and deploy LangChain/FastAPI apps	All
11	CI/CD with GitHub Actions for Python Projects	GitHub Docs + Example Pipelines	1 week	Automate tests, linting, deployment pipelines	All
12	Behavioral & Communication Skills	Internal HR Sessions	Ongoing	Improved documentation, communication, clarity	All
13	Open-Source Contribution or Personal AI Project	GitHub + Portfolio Building	Final Month	Showcase-ready project or contributions	All
	Final Project				
		End-to-end LLM-based app using LangChain + LangGraph + FastAPI			
		Working RAG pipeline using FAISS/Chroma with vector store			